

PDF Insights – AI Study Assistant

Problem Statement Document

Introduction

Students work with large volumes of lecture notes, textbooks, lab manuals, research papers, and academic PDFs. Locating specific information within these documents is often slow and inefficient.

Problem Statement

Students spend significant time searching through study materials to find answers, revise concepts, and prepare for examinations. Traditional keyword search is limited because it does not understand context. An intelligent study assistant is needed to answer questions directly from uploaded documents.

Proposed Solution

PDF Insights is a Retrieval-Augmented Generation (RAG) system that allows students to upload study PDFs and ask questions in natural language. Documents are processed, chunked, embedded, and stored in Pinecone Vector Database. Relevant content is retrieved and sent to a Large Language Model through Groq API to generate accurate, citation-backed answers.

Objectives

Upload PDFs, process study material, store embeddings in Pinecone, answer questions from uploaded documents, generate summaries, create quizzes and flashcards, and support efficient exam preparation.

Technology Stack

Frontend: React. Backend: FastAPI. Vector Database: Pinecone. LLM Provider: Groq API. Embeddings: BGE Small or Sentence Transformers. Database: PostgreSQL. Deployment: Docker.

Expected Outcome

Students can interact with their learning materials conversationally, receive accurate answers with references, generate revision notes, and improve learning efficiency.